KAIZEN MATHS

Lesson starter

Averages, Range and Missing Values

Comparing Measures

Name _____

S1

Which measure best represents this data? 22 25 29 31 57

S2

Compare outliers: 14 25 26 34 35 vs. 14 25 26 34 67

S3

Company salaries: $28,000 $42,000 $47,000 $51,000 $196,000 - Which measure for typical salary?

S4

Which measure best represents this data? 30 32 36 40 59

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Exit ticket

Averages, Range and Missing Values

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Name _____

E1

Compare outliers: 20 23 32 37 40 vs. 20 23 32 37 80

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Company salaries: $29,000 $34,000 $39,000 $50,000 $133,000 - Which measure for typical salary?

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Which measure best represents this data? 11 15 19 22 54

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Teacher answers & working

Averages, Range and Missing Values · Comparing Measures

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S1

Which measure best represents this data? 22 25 29 31 57

Answer:

Median (29) - less affected by the outlier

Working:

Mean = (22 + 25 + 29 + 31 + 57) ÷ 5 = 32.8

Median = 29

57 is much larger than the other values, so it pulls the mean higher.

The median is better because it is less affected by the outlier.

S2

Compare outliers: 14 25 26 34 35 vs. 14 25 26 34 67

Answer:

Mean changes more than median

Working:

Original mean = 26.8, median = 26

With outlier mean = 33.2, median = 26

The calculation shows: The median is resistant to outliers.

S3

Company salaries: $28,000 $42,000 $47,000 $51,000 $196,000 - Which measure for typical salary?

Answer:

Median ($47,000)

Working:

Mean = $72,800

Median = $47,000

The median better represents typical salary because the high salary skews the mean.

S4

Which measure best represents this data? 30 32 36 40 59

Answer:

Median (36) - less affected by the outlier

Working:

Mean = (30 + 32 + 36 + 40 + 59) ÷ 5 = 39.4

Median = 36

59 is much larger than the other values, so it pulls the mean higher.

The median is better because it is less affected by the outlier.

Teacher answers & working

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E1

Compare outliers: 20 23 32 37 40 vs. 20 23 32 37 80

Answer:

Mean changes more than median

Working:

Original mean = 30.4, median = 32

With outlier mean = 38.4, median = 32

The calculation shows: The median is resistant to outliers.

E2

Company salaries: $29,000 $34,000 $39,000 $50,000 $133,000 - Which measure for typical salary?

Answer:

Median ($39,000)

Working:

Mean = $57,000

Median = $39,000

The median better represents typical salary because the high salary skews the mean.

E3

Which measure best represents this data? 11 15 19 22 54

Answer:

Median (19) - less affected by the outlier

Working:

Mean = (11 + 15 + 19 + 22 + 54) ÷ 5 = 24.2

Median = 19

54 is much larger than the other values, so it pulls the mean higher.

The median is better because it is less affected by the outlier.